INTEGRAL CLOSURE COMMUTES WITH LOCALISATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Integrality transported along a ring map. Used to divide an integrality relation by powers of f (lines 7823-7829). Shows that the localised integral closure is contained in the integral closure of the localisation.
- Clearing denominators in an integrality relation over a localisation. Used in the converse direction (lines 7831-7842). Produces a single element of the multiplicative subset that makes the numerator integral over the source ring.

Lemma 0.1. *Integral closure commutes with localization: If $A \to B$ is a ring map, and $S \subset A$ is a multiplicative subset, then the integral closure of $S^{-1}A$ in $S^{-1}B$ is $S^{-1}B'$, where $B' \subset B$ is the integral closure of A in B.*

Proof. Since localization is exact we see that $S^{-1}B' \subset S^{-1}B$. Suppose $x \in B'$ and $f \in S$. Then $x^d + \sum_{i=1,\ldots,d} a_i x^{d-i} = 0$ in B for some $a_i \in A$. Hence also

$$(x/f)^d + \sum\nolimits_{i=1,\ldots,d} a_i/f^i (x/f)^{d-i} = 0$$

in $S^{-1}B$. In this way we see that $S^{-1}B'$ is contained in the integral closure of $S^{-1}A$ in $S^{-1}B$. Conversely, suppose that $x/f \in S^{-1}B$ is integral over $S^{-1}A$. Then we have

$$(x/f)^d + \sum\nolimits_{i=1,\ldots,d} (a_i/f_i)(x/f)^{d-i} = 0$$

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in $S^{-1}B$ for some $a_i \in A$ and $f_i \in S$. This means that

$$(f'f_1 \ldots f_d x)^d + \sum\nolimits_{i=1,\ldots,d} f^i (f')^i f_1^i \ldots f_i^{i-1} \ldots f_d^i a_i (f'f_1 \ldots f_d x)^{d-i} = 0$$

for a suitable $f' \in S$. Hence $f'f_1 \ldots f_d x \in B'$ and thus $x/f \in S^{-1}B'$ as desired. $\square$